

Daniel Wang

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EDUCATION

University of British Columbia

2028

Bachelor of Applied Science - Computer Engineering

Dean's List (2024–2025)

SKILLS

Languages Python, Java, JavaScript, C, C++, Verilog, MATLAB
Databases & Tools React, Pytest, Git, Jira, Agile, PostgreSQL, MySQL, Firestore
Embedded Systems STM32, RTOS, ADC, DMA, CAN Bus, UART

WORK EXPERIENCE

NETGEAR, Richmond, BC

May 2026 - Aug 2026

RF Software Developer Co-op

- Owned migration of 16 automated LTE Python test scripts from legacy R&S CMW500 to newer CMX500 test instrument, building adapter functions and SCPI command mappings for instrument configuration and measurement.
- Compared runtimes for RF tests across both platforms, finding CMX500 up to $2\times$ slower. Fixed redundant reattachment cycles during channel switching and a UI-write bottleneck via non-blocking execution.
- Ran test scripts to validate results against 3GPP spec, tracking bugs in Jira.
- Maintained and released three internal tools for RF team (automated test environment, automated calibration unit, and automated report generator), and added a Microsoft Teams web hook for receiving remote test pass/fail updates.

Vancouver Universal Medical Ltd., Richmond, BC

May 2025 - Aug 2025

Web Developer Intern

- Built and maintained the company website in WordPress for client bookings and business information.
- Wrote a Python script to pull online appointments to Google Sheets contact list and Google Calendar for scheduling.
- Automated client appointment reminders through iMessage, using appointment data pulled via the Google Calendar API.

STUDENT DESIGN TEAMS

Formula Racing SAE, University of British Columbia

Sept 2024 – Present

Electrical Subteam Member

Strain Gauge Data Acquisition Board

- Streamed real-time a-arm load data via CAN bus at 100 Hz on STM32, synchronized with vehicle AIM accelerometer data using ADC with DMA and STM32 HAL drivers for digital filtering.
- Flashed and debugged firmware via ST-Link, validating ADC readings and signal integrity.
- Soldered strain gauge amplifier and CIU PCBs. Validated output and voltage rails using oscilloscope and multimeter.
- Built a Python pipeline to calibrate and convert voltage to wheel forces and plotted strain gauge and accelerometer data.
- Built the wiring harness connecting the CIU to the car's electrical system to collect suspension load data on test days.

Vehicle System Validation Board (VSVB)

- Designed a benchtop debugger to power ECUs and modules for testing drive-by-wire, PDM, and dash off the car.
- Built programmable potentiometers, digital waveforms, and variable-reluctance waveforms to simulate analog, digital sensors and reluctors for crank/RPM.
- Implemented CAN message reading/writing and a USB software test environment for test case development.

PROJECTS

DrPocket.org

[Website](#)

IEEE Engineering Design Team Competition 2026, Best Presentation Award

- Built a React frontend with an interactive 3D body model (Three.js) for symptom selection and a chat and history panel.
- Built a serverless API endpoint using Vercel Functions that combines user symptom data, 3D model selections, and medical history to call Anthropic API, returning AI-generated health responses.
- Designed a PostgreSQL database and wrote SQL queries to store and retrieve medical history and session data.
- Implemented Firebase Authentication to manage secure user login and signup for persistent user accounts across sessions.